Gated and Grounded: An Agentic Multimodal RAG for Verifiable Question Answering on Scientific Papers

***Abstract*—Questions about scientific papers are often answered by tables and figures, yet single-pass multimodal retrieval-augmented generation (RAG) answers every question, supported or not, and gives the reader no way to check where a claim came from. In this paper, an agentic multimodal RAG system is presented that gathers text, tables and figures over several rounds, asks a sufficiency gate whether the gathered evidence is enough before any answer is written, verifies each claim against the evidence it cites, and draws a proof box on the page for every published claim. The system was built with an open 4-bit 7B vision-language model on a single consumer GPU and evaluated on 150 questions from SPIQA against a single-pass and a text-only baseline. Answers were graded by a model judge that agreed with a human grader on 22/25 answers (kappa 0.72). The agent was wrong on 43% of the questions, against 66% for single-pass, but it was also correct less often (21% against 34%), because its abstentions did not fall on the harder questions. When the same graph was run with a stronger model, abstention became selective. Where a claim cited the correct table or figure, its proof box pointed at the hand-annotated answer in 71% of cases, and every checked table-cell proof did. The gates are therefore shown to cut confident errors, while their full benefit depends on the capacity of the model that makes the judgements.**

***Index Terms*—retrieval-augmented generation, multimodal question answering, agentic retrieval, answer attribution, hallucination, scientific documents**

I. Introduction

A large share of what a scientific paper reports is never written out in its running text. The decisive number sits in a table cell, and the trend that supports a claim is drawn in a chart. A question answering system for papers therefore has to read tables and figures, and benchmarks such as SPIQA [1] were built for exactly this setting: each question is tied to the figure or table that holds its answer.

Most retrieval-augmented generation (RAG) systems [2] retrieve once and then answer. When the retrieved material does not contain the answer, the generator still produces one, and a fluent but unsupported answer is hard to tell apart from a correct one. A second problem is verification. Even a correct answer is of limited use to a researcher if the cell or chart region it rests on cannot be located, since the answer then has to be checked by reading the paper anyway.

In this work, both problems are approached with the same idea: the system is made to check before it answers. An agent gathers evidence over several rounds, choosing the next tool from a text, a table and a figure retriever. A sufficiency gate decides whether the evidence suffices before anything is written. Each claim of the draft must cite its evidence, and a faithfulness gate verifies every citation. Every published claim is then grounded by a box on the page: a table cell, a chart bar, a quoted sentence or a diagram label. When the evidence does not suffice, the system says so instead of guessing.

Three research questions were studied:

- **RQ1:** Does the agentic loop with its gates improve accuracy and reduce wrong answers compared with single-pass retrieval?
- **RQ2:** Can visual document retrieval bring the right tables and figures to the agent, and does seeing them help?
- **RQ3:** Can the agent ground its claims with precise boxes on the page?

The main contributions are as follows:

- An agentic multimodal RAG design with two gates and verified claim citations, implemented end to end with an open model on consumer hardware.
- An exact grounding scheme that boxes cells, bars, sentences and labels read from the PDF, with the retriever's similarity map kept only for photographs.
- A three-way evaluation (correct, wrong, abstained) with a checked model judge, an analysis of whether abstention is selective, and hand-annotated answer regions.
- Evidence, obtained by keeping the graph fixed and changing only the model, that the benefit of the gates is bounded by the model that makes their judgements.

Prior work on multimodal question answering, agentic retrieval and attribution is summarised in Section II. The agent and its two gates are detailed in Section III, and the data, baselines and metrics are listed in Section IV. The three research questions are answered in Section V, their implications are examined in Section VI, and future directions close the paper in Section VII.

II. Related Work

A. Multimodal Question Answering and Retrieval

Question answering over scientific papers was made a benchmark task by SPIQA [1], in which every question depends on a figure or table of the paper and is annotated with it. Such benchmarks score whether an answer is correct; the location of its evidence is not asked for, which leaves the verification problem open. For retrieval, ColPali [3] embeds page images as grids of patch vectors and scores them against the query by late interaction, so that pages can be found

without OCR. VisRAG [4] builds retrieval-augmented generation on such a retriever and passes the retrieved page images to a vision-language generator. In these systems retrieval is performed once, whereas in the present system pages and regions are retrieved repeatedly under the control of a planner.

B. Agentic and Self-Correcting RAG

Retrieval-augmented generation [2] conditions a generator on retrieved passages. In ReAct [5], reasoning steps and tool calls are interleaved, so that each observation shapes the next action, which is the pattern followed by the planner here. In Self-RAG [6], special reflection tokens are learned, so that a single model can judge whether retrieval is needed and grade what it has generated. The two gates used in this work play comparable roles, one judging the evidence before writing and one judging the claims after it, but they are realised by prompting an off-the-shelf model without any training, and they judge evidence from three modalities.

C. Attribution and Evaluation

ALCE [7] asks language models to cite the passages that support each statement and measures how well the citations hold. The claim citations used here follow the same idea, but every citation is verified before the claim is published. A language model is used as the judge of answer correctness, as studied in [8], and its verdicts are checked against a human grader. For visual attribution, the pointing game [9] tests whether the peak of a saliency map falls on the annotated object; the same test is applied in this work to proof boxes.

III. Proposed System

A. Overview

The system is built as a graph of nodes, each with a single task, in LangGraph. The flow of a question through the graph is shown in Fig. 1. The question is first analysed; the planner then selects one retrieval tool per round, and the results are added to an evidence buffer. The sufficiency gate (G1) either sends control back to the planner for another round or releases the buffer to the writer. The writer drafts claims with citations, the faithfulness gate (G2) checks every claim and returns failed drafts for rewriting, and the accepted claims are grounded with proof boxes before the answer is returned. Two loops are therefore present, and both are bounded: at most three retrieval rounds and at most two rewrites are allowed.

B. Question Analysis and Planning

The question is split into sub-goals, and the names it mentions (datasets, methods, metrics) are extracted for later filtering. The kind of evidence each sub-goal needs, a table or a figure, is not left to the language model. It is decided by a vote of the nine most similar labelled SPIQA questions taken from papers that are used in neither the development nor the test split. On the development questions, this vote chose the reference's kind in 17 of 20 cases, whereas the model alone chose it in 8. The planner then selects one tool per round and writes the query for it; a tool that has already been used for a sub-goal may be called again only with a new query.

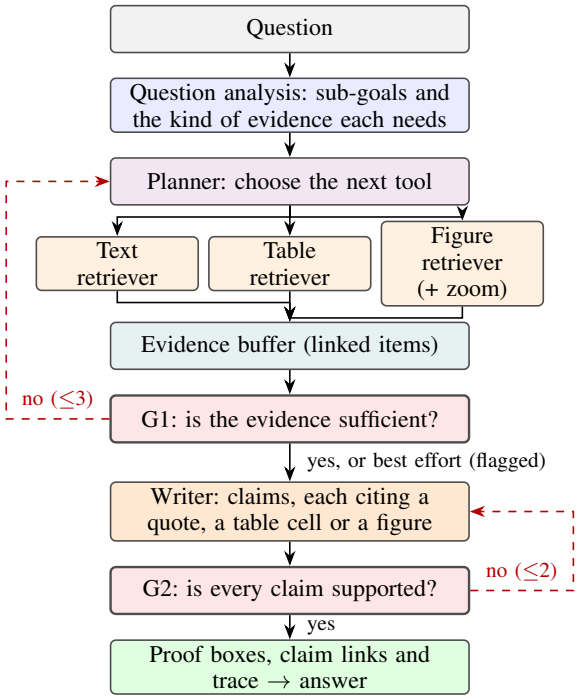


Fig. 1. Flow of a question through the system. Solid arrows show the forward path. Dashed arrows show the two bounded loops: the sufficiency gate (G1) returns control to the planner for another round (at most three), and the faithfulness gate (G2) returns the draft to the writer (at most two rewrites).

C. Retrieval Tools

Three tools can be selected by the planner. The text retriever ranks the paper's paragraphs by reciprocal rank fusion of two lists: SPECTER2 embeddings [10] searched with a FAISS index, and keyword overlap weighted by inverse document frequency. The keyword list was added because, within one paper, every paragraph shares the topic that the embedding captures, whereas SPIQA questions reuse the paper's own terms. The table retriever ranks the paper's tables by their caption words and by the visual score of their page, and returns each table as cells recovered with Table Transformer [11], each cell keeping its page coordinates. The figure retriever scores page images with ColQwen2, a model of the ColPali family [3], and returns the best figure. A fourth tool, a zoom that re-renders the figure at 300 dpi, is not chosen by the planner; it is run automatically when a figure covers less than 30% of its page. Every retrieved item enters the evidence buffer with its page, tool and round, and items are linked when they share a page or when a sentence names a table or figure.

D. Sufficiency Gate

Before anything is written, the buffer is judged by G1. The gate is asked to name the facts in the buffer that answer the question, with their evidence identifiers, and to give its verdict only afterwards. The order matters because the reply is generated as JSON in schema order: when the verdict came first, it was justified only after it had been made. If G1 is not convinced after three rounds, two cases are separated. When no answering fact was found, the system abstains and reports what is missing. When such facts were found but judged

incomplete, the answer is written from them as a *best effort* and is marked as possibly incomplete. In this way, G1 decides when the search stops, while the decision to publish a claim is left to G2.

E. Writing and Faithfulness Gate

The writer receives the buffer as text, cells and figure images, and returns a list of claims. Every claim must cite its evidence as a verbatim quote, a table cell given by row, column and value, or an observation about a figure. Each citation is verified by G2 in two layers. First, a quote must occur in the cited paragraph and a cell value must equal the parsed cell; both sides are normalised beforehand for the artefacts of PDF text extraction, such as ligatures, line-end hyphens and spaced decimals. Second, the model is asked whether the cited evidence supports the claim as written. A failed draft is returned with the reason, and claims that still fail after two rewrites are withheld and listed as unconfirmed.

F. Grounding the Claims

Every published claim is given a box on the page, and exact sources are preferred: the cited table cell, the bar of a vector chart read directly from the PDF, the quoted sentence located on the page, the caption, or the labels of a diagram that the claim names. Only for photographs, which carry no printed words, is the box taken from the highest-scoring patches of the ColQwen2 similarity map computed for the claim's text. An execution trace, listing the tools called and every gate decision, is returned with the answer.

IV. Experimental Setup

A. Data

Questions were drawn from SPIQA test-A [1] (118 papers, 666 questions) with a fixed seed. Whole papers were assigned to the splits, so that no paper contributes to both. A development split of 20 questions from 5 papers was used for every design decision. A test split of 150 questions from 31 papers (71 with a table and 79 with a figure as reference, at most 8 per paper) was used once. All settings were frozen before the test run, and the evaluation script refuses to run on the test split if any setting differs. A multi-hop subset of 10 test questions was selected by a fixed rule over the question text: it holds the questions that ask for a reason or an implication ("why", "suggest", "possible explanation") of a result shown in a table or figure.

B. Systems Compared

The same model, tools, prompts and grounding are used by every system, and each system removes one part of the design. *Full* is the complete agent. *Single-pass* has no planner, loops or gates: every tool is called once with the question, and the writer answers from the result, so that the effect of agency is isolated. *Text-only* is the full graph with the text tool alone, with tables and figures available only as extracted text, so that the effect of seeing them is isolated. *Strict G1* is the first frozen design, in which the agent abstains whenever G1 is not convinced (Section V-C). *Full (Gemini)* is the full graph with Gemini 3.5 Flash-Lite (free tier) behind every model call, so that the effect of the model is separated from that of the architecture.

C. Implementation

Every model role was filled by Qwen2.5-VL-7B-Instruct [12], run as a 4-bit quantised build behind a llama.cpp server. All experiments were run offline on one PC with an Intel Arc B580 GPU (12 GB) and 15.6 GB of RAM. Every model reply is constrained to a JSON schema, generated at temperature 0 with a fixed seed, and cached under a hash of the request, so that every answer can be reproduced exactly.

D. Outcomes and Judge

One of three outcomes is assigned to each question. It is *correct* or *wrong* when at least one claim reaches the reader, and *abstained* when the system reports that it could not confirm an answer or when every claim is withheld by G2. Answers were graded against SPIQA's gold answer by the local model, which was told to accept an answer that gives the main point of the gold answer and contradicts nothing in it, and to write its reason before its verdict. To check the judge, 30 answers (6 per system, shuffled) were graded by a human annotator who did not know the system or the judge's verdict.

E. Metrics

Besides the outcome rates and the precision when answering, four scores were computed. Retrieval faithfulness (RF) is the share of the distinct tables and figures in the buffer that are the reference, and the hit rate is the share of questions whose reference reached the buffer. The agent efficiency score (AES) is the number of hops needed (one, or two on the multi-hop subset) divided by the number of search calls, capped at one. For the cross-modal grounding score (CGS), each proof box of a question is laid over the hand-drawn answer boxes, and the largest ratio of their shared area to their combined area (IoU) is kept; a question left without a box scores zero. Since IoU was found to penalise correct proofs drawn at another size (Section V-E), the pointing game [9] is also reported: a proof counts as a hit when the centre of its box lies inside an answer box. This measure was added after the IoU results were seen. Finally, the share of published claims that cite the reference table or figure is reported as reference attribution. Uncertainty is shown as a 95% interval obtained by resampling the questions with replacement 10,000 times; when two systems are compared, the same resampled questions are used for both.

F. Annotation

For the grounding scores, 50 test questions (25 tables and 25 figures) were drawn at random before any output was inspected. On each page, the part of the table or figure holding the gold answer was boxed by a co-author with an offline annotation tool and written guidelines. In two quality-control

rounds, 16 boxes that covered a whole table or figure although the question named specific values were narrowed; they were selected by a fixed rule, without reference to any system's boxes.

V. Results

A. Overall Outcomes

The outcomes of the five systems on the 150 test questions are given in Table I, together with the precision when answering, that is, the share of answered questions that were answered correctly. The full agent answered 21% of the questions correctly, was wrong on 43% and abstained on 35%. Single-pass, which never abstains, was correct on 34% and wrong on 66%.

TABLE I
Outcomes on the 150 test questions in percent, with 95% bootstrap intervals in brackets, and precision when answering (Prec.)

System	Correct	Wrong	Abstained	Prec.
Full (7B)	21 [15, 28]	43 [35, 51]	35 [28, 43]	33
Single-pass	34 [27, 41]	66 [59, 73]	0 [0, 0]	34
Text-only	15 [10, 21]	51 [43, 59]	33 [26, 41]	23
Strict G1	5 [2, 9]	9 [5, 14]	85 [79, 91]	36
Full (Gemini)	30 [23, 37]	51 [43, 59]	19 [13, 26]	37

The same outcomes are drawn in Fig. 2, where the trade made by each system can be seen directly: the gated systems replace part of the wrong answers with abstentions, and the strict design replaces most answers of every kind.

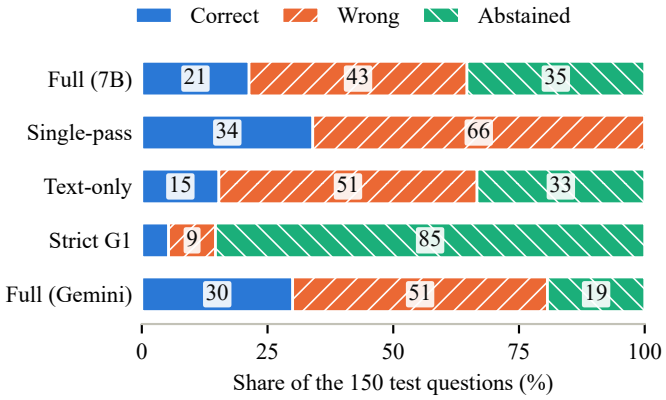


Fig. 2. Share of correct, wrong and abstained answers for each system on the 150 test questions. Hatching distinguishes the outcomes in grey-scale print.

The judge was found to be reliable. It agreed with the human grader on 22/25 answers (Cohen's kappa of 0.72), and 5 further answers were left undecided by the grader. In all three disagreements, the answer was accepted by the human and rejected by the judge, so the correct rates in Table I are, if anything, conservative.

B. RQ1: Agency

The paired differences between the systems are listed in Table II. Compared with single-pass, the full agent was wrong 22.7 percentage points less often, and the difference lies outside the bootstrap interval. It was also correct 12.7 points less often, because 35% of the questions were left unanswered. Its precision when answering (33%) was no higher than that of single-pass (34%), which shows that fewer wrong answers were obtained by answering less rather than by answering better.

TABLE II
Paired differences in percentage points on the same questions (*: the 95% bootstrap interval excludes zero; MH: multi-hop subset)

Comparison	Correct	Wrong	Abstained
Full − Single-pass	−12.7*	−22.7*	+35.3*
Full − Text-only	+6.0	−8.0	+2.0
Full − Strict G1	+16.0*	+34.0*	−50.0*
Gemini − Full (7B)	+8.7	+7.3	−16.0*
Full − Single-pass, MH	+20.0	−40.0*	+20.0
Full − Text-only, MH	+40.0*	+0.0	−40.0*

Whether the abstentions fall on the hard questions was tested with single-pass as a reference. A selective agent would hold back the questions that single-pass also gets wrong. The opposite was observed: single-pass was correct on 40% of the questions that the full agent held back, but on only 31% of those it answered. The best-effort flag carried little information as well, since flagged answers were correct about as often as those accepted by G1 (24 of 75 against 8 of 22). On the multi-hop subset, where the loop was expected to matter most, the full agent was wrong on 2 of the 10 questions against 6 for single-pass, but ten questions allow only an indicative reading. The search tools were called 2.7 times per question by the agent and 3 times by single-pass, which gives a higher efficiency score (AES 0.44 against 0.36). The planner and the gates, however, raised the number of model calls from 3 to 11.7 per question.

C. A Change Made During Evaluation

In its first frozen form, the agent abstained whenever G1 was not convinced after three rounds. In the first test run, 110 of the first 142 questions were abstained before a power cut stopped the run, although the reference was in the buffer for about two thirds of them. The best-effort rule of Section III was then designed on the development split only, without any test answer being read; on that split it moved the agent from 6 right, 1 partly right, 0 wrong and 13 abstained to 9, 4, 5 and 2. All systems were re-run on the test split, and the first design was kept as the strict G1 ablation.

D. RQ2: Visual Retrieval

The reference table or figure reached the buffer for 73% of the questions with the full agent and for 77% with single-pass, which fetches every kind of evidence in every run. Of the distinct tables and figures fetched by the agent, 44% were the reference (RF 0.44, against 0.41 for single-pass and 0.55 with Gemini). On the development split, ColQwen2 ranked the reference's page first for 15 of 20 questions. Compared with the text-only agent, the full agent was correct 6 points more

often and wrong 8 points less often (Table II), and neither difference lies outside its interval. On the multi-hop subset, however, the full agent was correct 40 points more often.

E. RQ3: Grounding

The grounding scores on the 50 annotated questions are given in Table III. Measured by IoU, grounding was poor for every system. On inspection, a correct proof was often scored low because it was drawn at a different size: a parsed cell spans the width of its column, while the hand box was drawn tightly around the printed number, so that two exactly right cells scored IoU 0.43 and 0.46. Shrinking each cited cell to its printed words did not raise CGS (0.070 against 0.068), as many questions lost as gained. The pointing game removes this effect, and it credits the full agent on 0.28 of its answered questions.

TABLE III
GROUNDING ON THE 50 ANNOTATED QUESTIONS (CGS: BEST IOU; POINT.: POINTING GAME, ALL AND ANSWERED QUESTIONS) AND REFERENCE ATTRIBUTION ON ALL 150 QUESTIONS

System	CGS	Point.	Point. (ans.)	Ref. attr.
Full (7B)	0.07	0.16	0.28	0.44
Single-pass	0.15	0.40	0.40	0.42
Text-only	0.00	0.04	0.07	0.00
Full (Gemini)	0.05	0.20	0.26	0.55

The low scores were traced to the choice of sources rather than to the boxes. Of the 72 proof boxes drawn by the full agent on the annotated questions, only 31 lay on the reference table or figure; the others pointed at paragraphs or at other tables. Of the boxes on the reference, 71% pointed inside the hand-drawn answer box, and every table-cell proof did (12/12). For single-pass, whose cells are not checked by a gate, 30/44 cell proofs were inside.

F. A Stronger Model behind the Same Graph

With Gemini behind every call and nothing else changed, the agent was correct on 30%, wrong on 51% and abstained on 19%. G1 was convinced on 117 questions instead of 22, after 1.9 rounds on average instead of 2.7, and G2 kept 88% of the drafted claims instead of 50%. Most importantly, its abstention was selective: single-pass was correct on only 14% of the questions it held back, against 39% of those it answered.

VI. DISCUSSION

A. The Model Limits the Gates

The design rests on the expectation that a system which checks its evidence and its claims would turn confident errors into stated gaps. With the 7B model, only half of this expectation was met: wrong answers became far less frequent, but easy questions were held back as readily as hard ones. Because the graph, prompts and settings were kept identical when the model was exchanged, the two explanations can be separated. With the stronger model, G1 was convinced about five times as often, far fewer claims were rejected by G2, and the held-back questions were the hard ones. The gates therefore behave as intended when the model behind them can judge sufficiency and read tables and charts reliably, and a 4-bit 7B model cannot yet do so consistently. A practical arrangement follows from this result: a small local model can read and write, while the two judgements are delegated to a stronger model.

B. Measuring Grounding

IoU against a hand-drawn box was fixed as the grounding measure before the evaluation, but it was found to compare two drawing conventions as much as the precision of a proof. A cell box spans its column, while an annotator boxes the number, or a whole row for a comparison. The pointing game is insensitive to these conventions and is therefore reported next to IoU. In future annotation, the unit should be fixed in advance, for instance whole cells for tables.

C. Threats to Validity

The judge is the same model as the evaluated local systems; its agreement with a human grader was substantial, but it was checked on 25 answers only. The answer boxes were drawn by one annotator who, as a co-author, knew the aims of the work. The best-effort rule was introduced after a first test run, and the pointing game after the IoU results; both are disclosed, and the original measures are reported. The study covers 150 questions from 31 computer-science papers, retrieval within a known paper, one quantised model run once at temperature 0, and a multi-hop subset of ten questions. Reference attribution, used in place of a hand check of each citation, undercounts claims that correctly cite supporting text. Finally, the benefit of proof boxes for human verification was argued from their precision and was not measured in a user study.

VII. CONCLUSION

An agentic multimodal RAG system was presented that gathers evidence over several rounds, checks the sufficiency of the evidence before writing, verifies every claim against its citation, and grounds each published claim with a box on the page. On 150 SPIQA questions, wrong answers were reduced from 66% to 43% compared with single-pass retrieval, at the cost of abstentions that, with a 7B model, did not single out the hard questions; with a stronger model behind the same graph, abstention became selective. Where a claim cited the right table or figure, its proof box pointed at the answer in 71% of cases. In future work, a hybrid in which a small model reads and writes while a stronger model makes the two judgements will be evaluated on unused SPIQA papers, claims will be steered towards the table or figure that holds the answer, and the value of the proofs for human verification will be measured in a user study.

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